Lab 08

This lab builds upon the database schema created in **Lab 07** by introducing database manipulation using **Java JDBC** and covering advanced SQL concepts like **Views** and **Indexes**.

Employee

emp_id	name	salary	address_id	dept_id
111	Zaineh	100000	1	1
112	Yasmeen	160000	2	4
113	Mira	140000	3	3
114	Shimaa	200000	4	2
115	Dean	150000	5	1

Address

address_id	city	state	zipcode
1	Fairfield	IA	52556
2	Iowa City	IA	52440
3	Morrison	IL	61270
4	Orlando	FL	34565
5	Tampa	FL	31765

Department

dept_id	name
1	Tech
2	HR
3	Finance
4	Marketing

Employee_Project

emp_id	project_id
115	1
115	2
115	3
114	1
114	3
111	1
111	2

Project

project_id	project_name	estimated_days	location
1	X	180	FL
2	Y	60	FL
3	Z	80	IA

Advanced SQL Tasks

Before implementing the Java class, perform the following SQL tasks on your existing database from Lab 07.

A. Creating a Database View

Create a **VIEW** named `EmployeeDetails` that joins the `Employee`, `Department`, and `Project` tables. The view should display the following columns:

- `emp_id` (from Employee)
- `employee_name` (from Employee, renamed to be distinct)
- `salary`
- `dept_name` (from Department)
- `project_name` (from Project)

B. Adding an Index

Add an **INDEX** named `idx_employee_name` to the `name` column of the `Employee` table.

Java Repository Implementation (JDBC)

A. Create Model Class (`Employee.java`)

You must first create the simple Java model class `Employee` to hold the data retrieved from or sent to the database.

B. Create Repository Class (`EmployeeRepository.java`)

Create a Java class named `EmployeeRepository`. This class must implement the following five core CRUD methods for interacting with the `Employee` table. Include the correct SQL for each operation.

Method Signature	Operation
`List<Employee> findAll()`	Gets all employees.
`Employee findById(int empId)`	Gets an employee.
`void create(Employee employee)`	Inserts a new employee.
`void update(Employee employee)`	Updates an employee
`void delete(int empId)`	Deletes an employee

Add a **`main` method** to the `EmployeeRepository` class. The `main` method should execute each of the five methods sequentially to demonstrate a complete CRUD lifecycle:

1. Instantiate `EmployeeRepository`.
2. Create a new `Employee` object and execute `create()`.
3. Execute `findAll()` to show the newly added employee.
4. Execute `findById()` to retrieve the specific employee.
5. Update the employee object and execute `update()`.
6. Execute `delete()`.
7. Execute `findAll()` again to confirm the deletion.

Submission:

You should submit your completed EmployeeRepository.java file, which includes the demonstration main method. Additionally, include screenshots of the created database objects: specifically, show the definition or schema details for the EmployeeDetails view and visual proof that the idx_employee_name index has been applied to the Employee table.